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TERMINAL CAP ASSEMBLY AND BATTERY MODULE INCLUDING TERMINAL CAP ASSEMBLY

Abstract

A battery module includes a plurality of sub-modules including a first sub-module and a second sub-module respectively including a plurality of battery cells, a first terminal portion electrically connected to the plurality of battery cells of the first sub-module and including one or more electrode terminals including a first electrode terminal, a second terminal portion electrically connected to the plurality of battery cells of the second module and including one or more electrode terminals including a first electrode terminal, and a terminal cap assembly covering the first terminal portion and the second terminal portion and electrically connecting the first terminal portion and the second terminal portion to each other. The terminal cap assembly includes a fuse wire and a protector covering the fuse wire, the first electrode terminal of the first terminal, and the first electrode terminal of the second terminal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This patent document claims the priority and benefits of Korean Patent Application No. 10-2024-0030520 filed on Mar. 4, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The disclosure and implementations disclosed in this patent document generally relate to a terminal cap assembly and a battery module including the same.

BACKGROUND

[0003] Secondary unlike primary batteries, are convenient, in that the secondary batteries can be charged and discharged, and thus have drawn much attention as power sources for various mobile devices and electric vehicles.

[0004] Such a secondary battery may include a battery cell. An electrode assembly, formed by stacking a cathode plate, an anode plate, and a separator or winding the cathode plate, the anode plate, and the separator in the form of a roll, may be accommodated in a case of the battery cell. A plurality of battery cells may be stacked in a predetermined direction and accommodated in a battery module or a battery pack.

[0005] Such a battery module or battery pack may have a risk of thermal runaway due to factors such as the occurrence of a short-circuit or overcurrent. Accordingly, there is a need for research into a structure for protecting a battery module or a battery pack from factors such as the occurrence of a short-circuit or overcurrent.

SUMMARY

[0006] A terminal cap assembly and a battery module including the same of the present disclosure may prevent a short-circuit of the battery module and cut off an electrical connection between a plurality of sub-modules when an overcurrent occurs, thereby ensuring safety of the battery module.

[0007] The terminal cap assembly and the battery module including the same of the present disclosure may be fixed to terminals of two adjacent sub-modules to fix the two adjacent sub-modules.

[0008] The terminal cap assembly and the battery module including the same of the present disclosure may minimize the influence of various environmental factors that may hinder safety during transportation, thereby improving safety.

[0009] The battery module of the present disclosure may be widely applied in the field of green technology, such as to electric vehicles, battery charging stations, and other battery-utilizing solar power generation schemes, wind power generation schemes, or the like. In addition, the battery module including a pad assembly of the present disclosure may be used in eco-friendly electric vehicles, hybrid vehicles, or the like, to prevent climate change by suppressing air pollution and greenhouse gas emissions.

[0010] According to an aspect of the present disclosure, there is provided a battery module including a plurality of sub-modules including a first sub-module and a second sub-module respectively including a plurality of battery cells, a first terminal portion electrically connected to the plurality of battery cells of the first sub-module, the first terminal portion including one or more electrode terminals including a first electrode terminal, a second terminal portion electrically connected to the plurality of battery cells of the second module, the second terminal portion including one or more electrode terminals including a first electrode terminal, and a terminal cap assembly covering the first terminal portion and the second terminal portion, the terminal cap assembly electrically connecting the first terminal portion and the second terminal portion to each other. The terminal cap assembly may include a fuse wire electrically connecting the first electrode terminal of the first terminal and the first electrode terminal of the second terminal to each other, and a protector covering the fuse wire, the first electrode terminal of the first terminal, and the first electrode terminal of the second terminal.

[0011] The plurality of battery cells may be stacked in a first direction. The battery module may further include a cap cover covering the electrode terminal that is not covered by the terminal cap assembly.

[0012] The first terminal portion may further include a second electrode terminal spaced apart from the first electrode terminal of the first terminal portion in the first direction, and the second terminal portion may further include a second electrode terminal spaced apart from the first electrode terminal of the second terminal portion in the first direction. The cap cover may independently cover the second electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion.

[0013] The first electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion may have different electrical polarities. A terminal cap assembly may be disposed at one sides of the plurality of sub-modules in the first direction, and a cap cover may be disposed at the other sides of the plurality of sub-modules in the first direction.

[0014] The terminal cap assembly may be provided as one terminal cap assembly and connected to the first electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion. The terminal cap may be provided as two terminal caps, and the two terminal caps may be disposed on the second electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion, respectively.

[0015] The fuse wire may be provided to fracture due to a current having a predetermined value or more, and to cut off an electrical connection between the first terminal portion and the second terminal portion.

[0016] The fuse wire may have a diameter of 0.25 mm.sup.2 to 1 mm.sup.2.

[0017] The protector may include a first terminal cover covering the first electrode terminal of the first terminal portion, a second terminal cover covering the first electrode terminal of the second terminal portion, and a connection cover connecting the first terminal cover and the second terminal cover to each other.

[0018] The electrode terminal of the first terminal portion and the electrode terminal of the second terminal portion may have terminal holes, respectively. The first terminal cover and the second terminal cover may include a first protruding body and a second protruding body protruding to be inserted into the terminal holes, respectively.

[0019] The fuse wire may include a first connection portion and a second connection portion in contact with and electrically connected to the first electrode terminal of the first terminal portion and the first electrode terminal of the second terminal portion, respectively. The first protruding body and the second protruding body may pass through the first connection portion and the second connection portion, and may be inserted into the first electrode terminal of the first terminal portion and the first electrode terminal of the second terminal portion, respectively.

[0020] The protector may include a first seating body and a second seating body respectively provided at one sides of the first protruding body and the second protruding body. In a state in which the terminal cap assembly is coupled to the first terminal portion and the second terminal portion, the first connection portion may be disposed between the first seating body and the first terminal portion, and the second connection portion may be disposed between the second seating body and the second terminal portion.

[0021] The fuse wire may further include a connector connecting the first connection portion and the second connection portion to each other, and at least a portion of the connector may be inserted into the connection cover, and may not be externally exposed.

[0022] The battery module may further include a first cover support portion and a second cover support portion protruding from edges of the first and second terminal portions, the first cover support portion and the second cover support portion forming a seating space in which the terminal cap assembly is to be mounted. The terminal cap assembly may be mounted in the seating space to cover the first electrode terminal of the first terminal portion and the first electrode terminal of the second terminal portion.

[0023] The first terminal cover and the second terminal cover may respectively include a first cover body and a second cover body disposed to respectively oppose the first terminal portion and the second terminal portion in the first direction, and a first support body and a second support body respectively extending from one sides of the first cover body and the second cover body in the first direction, the first support body and the second support body respectively opposing the first terminal portion and the second terminal portion in a second direction, perpendicular to the first direction.

[0024] According to another aspect of the present disclosure, there is provided a battery module including a plurality of sub-modules including a first sub-module and a second sub-module respectively including a plurality of battery cells stacked in a first direction, a first terminal portion electrically connected to the plurality of battery cells of the first sub-module, the first terminal portion including a plurality of electrode terminals, a second terminal portion electrically connected to the plurality of battery cells of the second sub-module, the second terminal portion including a plurality of electrode terminals, a terminal cap assembly electrically connecting one electrode terminal, among the plurality of electrode terminals of the first terminal portion and one electrode terminal, among the plurality of electrode terminals of the second terminal portion, to each other, the terminal cap assembly covering the one electrode terminal of the first terminal portion and the one terminal of the second terminal portion together, and a plurality of cap covers independently covering remaining electrode terminals, among the plurality of electrode terminals of the first terminal portion, and remaining electrode terminals, among the plurality of electrode terminals of the second terminal portion, respectively. The terminal cap assembly may be disposed at one sides of the plurality of sub-modules in the first direction, and the plurality of cap covers may be disposed at the other sides of the plurality of sub-modules in the first direction.

[0025] According to another aspect of the present disclosure, there is provided a terminal cap assembly including a fuse wire electrically connecting a first terminal portion and a second terminal portion, respectively electrically connected to a plurality of different battery cells, to each other, and a protector covering at least a portion of the fuse wire, the first terminal portion, and the second terminal portion, the protector coupled to the first terminal portion and the second terminal portion.

[0026] The fuse wire may be provided between the first terminal portion and the second terminal portion to fracture due to a current having a predetermined value or more, and to cut off an electrical connection between the first terminal portion and the second terminal portion.

[0027] The fuse wire may include a first connection portion and a second connection portion respectively in contact with the first terminal portion and the second terminal portion, and a connector electrically connecting the first connection portion and the second connection portion to each other.

[0028] The protector may include a first terminal cover and a second terminal cover respectively covering the first terminal portion and the second terminal portion, and a connection cover disposed between the first terminal cover and the second terminal cover, the connection cover surrounding the connector.

[0029] The first terminal cover and the second terminal cover may respectively include a first

protruding body and a second protruding body respectively passing through the first connection portion and second connection portion, the first protruding body and the second protruding body respectively inserted into through-holes of the first terminal portion and the second terminal portion.
[0030] Configurations according to the aspects of the present disclosure have been described above, but the configurations are exemplary. Even when other configurations, not set forth herein, are added, it should be understood that the other configurations are within the scope of the present disclosure.
[0031] According to an embodiment of the present disclosure, a terminal cap assembly and a battery module including the same may cut off an electrical connection between a plurality of sub-modules when an overcurrent occurs, thereby ensuring safety.
[0032] The terminal cap assembly and the battery module including the same may be fixed to terminals of two adjacent sub-modules to fix the two adjacent sub-modules.
[0033] The terminal cap assembly and the battery module including the same may minimize the influence of various environmental factors that may hinder safety during transportation, thereby improving safety.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0034] Certain aspects, features, and advantages of the present disclosure are illustrated by the following detailed description with reference to the accompanying drawings.
[0035] FIG. **1** is a perspective view of a battery module according to an embodiment of the present disclosure.
[0036] FIG. **2** is an exploded perspective view of a battery module according to an embodiment of the present disclosure.
[0037] FIG. **3** is an exploded perspective view of a first sub-module and a second sub-module.
[0038] FIG. **4** is an enlarged view of portion “A” of FIG. **1**.
[0039] FIG. **5** is a view of FIG. **4** from which a terminal cap assembly is removed.
[0040] FIG. **6** is a perspective view of a terminal cap assembly according to an embodiment of the present disclosure.
[0041] FIG. **7** is a view of FIG. **6** in a different direction.
[0042] FIG. **8** is a bottom view of a terminal cap assembly according to an embodiment.
[0043] FIG. **9** is a side view of a terminal cap assembly according to an embodiment.
DETAILED DESCRIPTION
[0044] Before describing embodiments of the present disclosure, the words and terminologies used in the specification and claims should not be construed with common or dictionary meanings, but construed as meanings and conception coinciding the spirit of the present disclosure under a principle that the inventor(s) may appropriately define the conception of the terminologies to explain the present disclosure in the optimum method.
[0045] The same reference numerals in the drawings refer to components or elements performing substantially the same function. For ease of description and understanding, the same reference numerals may be used in different embodiments.
[0046] As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this disclosure, specify the presence of stated features, integers, operations, operations, elements, components or a combination thereof, but do not preclude the presence or addition of one or more other features,

integers, operations, operations, elements, components, and/or groups thereof.
[0047] In addition, the terms “upper side,” “upper portion,” lower side, ““lower portion,” “side surface,” “front surface,” “rear surface” and the like are described based on a direction illustrated the drawings, and may be described differently when a direction of a corresponding object is changed.
[0048] In addition, as used herein, terms including an ordinal number such as “first” and “second” may be used to distinguish between components. The ordinal number is used to distinguish the same or similar components from each other, and the meaning of the term should not be limitedly interpreted due to the use of the ordinal number. For example, components combined with the ordinal number should not be construed as limiting the order of use or arrangement by the number. If necessary, respective ordinal numbers may be used interchangeably.
[0049] Hereinafter, the present disclosure will be described in detail with reference to the accompanying drawings. However, embodiments are merely exemplary, and the present disclosure is not limited to specific embodiments.
[0050] A battery module **10**, on which a terminal cap assembly **50** is mounted, of the present disclosure will be described first.
[0051] FIG. **1** is a perspective view of a battery module according to an embodiment of the present disclosure. FIG. **2** is an exploded perspective view of a battery module according to an embodiment of the present disclosure. FIG. **3** is an exploded perspective view of a first sub-module and a second sub-module. FIG. **4** is an enlarged view of portion “A” of FIG. **1**. FIG. **5** is a view of FIG. **4** from which a terminal cap assembly is removed.
[0052] Referring to FIGS. **1** to **5** together, the battery module **10** of the present disclosure may include a sub-module **100** including a first sub-module **100***a* and a second sub-module **100***b* respectively including a plurality of battery cells **111**. The sub-module **100** may be accommodated in a module case **11** forming the exterior of the battery module **10**.
[0053] The module case **11** may include a first plate **12** (an upper cover) disposed above the sub-module **100** to cover an upper portion of the sub-module **100**, a second plate **13** including a lower cover **13***a* supporting a lower portion of the sub-module **100**, and a side cover **13***b* disposed on a side of the sub-module **100** to be parallel to the plurality of battery cells **111**, and an end plate **14** disposed on both sides of the sub-module **100** in a direction, opposing the first sub-module **100***a* and the second sub-module **100***b*.
[0054] A center plate **15** may be disposed between the first sub-module **100***a* and the second sub-module **100***b* to support two adjacent sub-modules **100**.
[0055] The upper cover **12** may have a terminal hole **125** to expose a terminal portion **135** of the sub-module **100** to be described below. The terminal portion **135** may be exposed to the outside of the module case **11** through the terminal hole **125**, and thus a terminal cap assembly **50** to be described below may be mounted on the terminal portion **135** (see FIGS. **4** and **5**).
[0056] A plurality of sub-modules **100** may include a cell assembly **110** formed by stacking the plurality of battery cells **111**, an insulating cover **120** disposed to oppose the end plate **14**, and a busbar assembly **130** electrically connected to the plurality of battery cells **111**.
[0057] The cell assembly **110** may include the plurality of battery cells **111** stacked to be parallel to each other in a predetermined direction. A direction in which the plurality of battery cells **111** are stacked and a direction in which the sub-modules **100** are disposed may be perpendicular to each other.
[0058] The plurality of battery cells **111** may convert chemical energy into electrical energy and supply power to an external circuit, or may receive power externally and convert electrical energy into chemical energy to store electricity. For example, the battery cell **111** may include a nickel metal hydride (Ni-MH) battery or a lithium ion (Li-ion) battery capable of being charged and discharged.
[0059] Each of the plurality of battery cells **1000** may accommodate an electrode assembly (not

illustrated) formed by stacking a cathode plate and an anode plate. In the electrode assembly, the cathode plate and the anode plate may be stacked with a separator interposed therebetween in a state in which wide surfaces of the cathode plate and the anode plate oppose each other. The separator may prevent an electrical short-circuit between the cathode plate and the anode plate, and may generate a flow of ions. For example, the separator may include a porous polymer film or a porous nonwoven fabric.

[0060] In addition, the electrode assembly may be a jelly-roll type electrode assembly formed by winding the cathode plate, the anode plate, and the separator in a predetermined direction, and various-types of electrode assemblies, such as a stacking type electrode assembly, a Z-folding type electrode assembly, and a stack-folding type electrode assembly, may be accommodated in a case.

[0061] The plurality of battery cells **111** may be a pouch-type secondary battery, a prismatic-type secondary battery, or a cylindrical-type secondary battery depending on a structure of the case.

[0062] The insulating cover **120** may include an electrically insulating material to prevent a short-circuit between the cell assembly **110** and the end plate **14**.

[0063] The busbar assembly **130** may include a busbar electrically connecting the plurality of battery cells **111** to each other, and a support frame (not illustrated) supporting the busbar. In addition, the busbar assembly **130** may be formed such that the terminal portion **135** is disposed to be close to an adjacent sub-module **100**. That is, the busbar may electrically connect the plurality of battery cells **111** and the terminal portion **135** to each other, and the terminal portion **135** may be electrically connected to an external power source.

[0064] The first sub-module **100***a* and the second sub-module **100***b* may include cell assemblies (a first cell assembly **110***a* and a second cell assembly **110***b*), insulating covers **120***a* and **120***b*, busbar assemblies **130***a* and **130***b*, and terminal portions (a first terminal portion **135***a* and a second terminal portion **135***b*).

[0065] The busbar assemblies **130***a* and **130***b* of the first sub-module and the second sub-module **100***a* and **100***b* may include first busbar assemblies **131***a* and **131***b* disposed at one sides of the cell assemblies **110***a* and **110***b* to oppose the center plate **15**, and second busbar assemblies **132***a* and **132***b* disposed at the other sides of the cell assemblies **110***a* and **110***b* to oppose the end plate **14**.

[0066] The terminal portions **135***a* and **135***b* may be disposed to be close to a plurality of sub-modules **100***a* and **100***b*. For example, the terminal portions **135***a* and **135***b* may be provided in the first bus bar assemblies **131***a* and **131***b*.

[0067] A terminal portion (first terminal portion) **135***a* of the first sub-module **100***a* and a terminal portion (second terminal portion) **135***b* of the second sub-module **100***b* may be spaced apart from each other in a direction (X-axis direction) in which the cell assemblies **100***a* and **100***b* are stacked, and may respectively one or more electrode terminals.

[0068] For example, the one or more electrode terminals may respectively include first electrode terminals **1351***a* and **1351***b* disposed to close to one sides thereof in the direction (X-axis direction) in which the cell assemblies **100***a* and **100***b* are stacked, and second electrode terminals **1352***a* and **1352***b* disposed to close to the other sides thereof in the direction (X-axis direction) in which the cell assemblies **100***a* and **100***b* are stacked.

[0069] A first electrode terminal **1351***a* of the first terminal portion **135***a* and a first electrode terminal **1351***b* of the second terminal portion **135***b* may be disposed to oppose each other in a direction (Y-axis direction) in which the sub-modules **100** are arranged, and a second electrode terminal **1352***a* of the first terminal portion **135***a* and a second electrode terminal **1352***b* of the second terminal portion **135***b* may be disposed to oppose each other in the direction (Y-axis direction) in which the sub-modules **100** are arranged.

[0070] In addition, the first electrode terminals **1351***a* and **1351***b* and the second electrode terminals **1352***a* and **1352***b* of the terminal portion **135** may have opposite electrical polarities. Electrical polarities of the first electrode terminal **1351***a* and the second electrode terminal **1352***a* of the first terminal portion **135***a* may be opposite to electrical polarities of the first electrode terminal **1351***b* and the second electrode terminal **1352***b* of the second terminal portion **135***b*, respectively. For example, the first electrode terminal **1351***a* of the first terminal portion **135***a* and the second electrode terminal **1352***b* of the second terminal portion **135***b* may be cathodes, and the second electrode terminal **1352***a* of the first terminal portion **135***a* and the first electrode terminal **1351***b* of the second terminal portion **135***b* may be anodes.

[0071] The insulating covers **120***a* and **120***b* of the first sub-module and the second sub-module **100***a* and **100***b* may include first insulating covers **121***a* and **121***b* disposed between the cell assemblies **110***a* and **110***b* and the center plate **15**, and second insulating covers **122***a* and **122***b* disposed between the cell assemblies **110***a* and **110***b* and the end plate **14**, respectively. The first insulating covers **121***a* and **121***b* may prevent a short-circuit between the first bus bar assemblies **131***a* and **131***b* and the center plate **15**, and the second insulating covers **122***a* and **122***b* may prevent a short-circuit between the second bus bar assemblies **132***a* and **132***b* and the end plate **14**. In addition, the first insulating covers **121***a* and **121***b* may include a cover support portion **136** (to be described below) protruding to surround the terminal portion **135**.

[0072] In the present disclosure, among the plurality of sub-modules **100**, the first sub-module **100***a* and the second sub-module **100***b* may be disposed to be symmetrical to each other with respect to the center plate **15**, and may have the same configuration.

[0073] Accordingly, as described above, for ease, a configuration of the first sub-module **100***a* may be referred to as a first configuration, and a configuration of the second sub-module **100***b* may be referred to as a second configuration. For example, among the terminal portions **135**, the terminal portion **135***a* of the first sub-module **100***a* may be referred to as the first terminal portion **135***a*, and the terminal portion **135***b* of the second sub-module **100***b* may be referred to as the second terminal portion **135***b*.

[0074] In addition, as described above, even with the same configuration, reference numeral "˜a" may refer to a configuration of the first sub-module **100***a*, and reference numeral "˜b" may refer to a configuration of the second sub-module. For example, the first electrode terminal **1351***a* may refer to a configuration of the first terminal portion **135***a*, and the first electrode terminal **1351***b* may refer to a configuration of the second terminal portion **135***b*.

[0075] The first terminal portion **135***a* and the second terminal portion **135***b* may oppose each other with the center plate **15** interposed therebetween. That is, the terminal portions **135** of the plurality of sub-modules **100** may be disposed to be close to each other, not on the outside of the battery module **10**. Referring to FIGS. **4** and **5**, the terminal cap assembly **50** may be mounted on the terminal portion **135** exposed through the terminal hole **125**.

[0076] In embodiments of the present disclosure, the busbar assembly **130** may further include the cover support portion **136** protruding along a periphery of the terminal portion **135** to form a seating space C (see FIG. **4**). At least a portion of the cap assembly **50** may be seated in the seating space C, such that the cap assembly **50** may be mounted on the battery module **10**.

[0077] Specifically, the cover support portion **136** may be formed of an electrically insulating material to prevent the terminal portion **135** from being connected to an external conductor material. The cover support portion **136** may protrude upwardly from an edge of the terminal portion **135** to form the seating space C. At least a portion of the terminal cap assembly **50** may be accommodated in the seating space C formed by the cover support portion **136**, such that the terminal cap assembly **50** may be stably mounted on the terminal portion **135**.

[0078] As described above, the cover support portion **136** may include a first cover support portion **136***a* provided in the first terminal portion **135***a*, and a second cover support portion **136***b* provided in the second terminal portion **135***b*.

[0079] In addition, for ease, the seating space C, formed by the first cover support portion **136***a*, may be referred to as a first seating space, and the seating space C, formed by the second cover support portion **136***b*, may be referred to as a second seating space.

[0080] In an embodiment of the present disclosure, during transportation of the battery module **10**, conductive particles may be likely to come into contact with the terminal portion **135**, externally exposed through the terminal hole **125**, to cause a short-circuit. Accordingly, the terminal cap assembly **50** and the cap cover **60** to be described below may cover the terminal portion **135** to prevent a short-circuit.

[0081] An overcurrent may occur due to impacts, internal defects, or the like of at least one of the plurality of sub-modules **100** (for example, at least one of the first sub-module **100***a* and the second sub-module **100***b*).

[0082] Accordingly, the terminal cap assembly **50** of the present disclosure may include a protector **53** fixed to the terminal portion **135** to cover the terminal portion **135**, and a fuse wire **51** provided in the protector **53** to electrically connect adjacent terminal portions **135***a* and **135***b* to each other, the fuse wire **51** provided to fracture when a predetermined current (for example, an overcurrent having a predetermined value or more) flows (see FIG. **7**). That is, the terminal cap assembly **50** may electrically connect the plurality of sub-modules **100** to each other through the fuse wire **51**. The cap cover **60** may independently cover the electrode terminals **1351** and **1352** of the terminal portion **135** regardless of an electrical connection relationship.

[0083] According to an embodiment, the first sub-module **100***a* and the second sub-module **100***b* may be electrically connected to each other by one terminal cap assembly **50** disposed on one of the first electrode terminals **1351***a* and **1351***b* and the second electrode terminals **1352***a* and **1352***b*, and the cap cover **60** may be disposed on remaining electrode terminals **1351** and **1352**. For example, the terminal cap assembly **50** may electrically connect the first electrode terminal **1351***a* of the first terminal portion **135***a* and the first electrode terminal **1351***b* of the second terminal portion **135***b* to each other while covering the first electrode terminal **1351***a* of the first terminal portion **135***a* and the first electrode terminal **1351***b* of the second terminal portion **135***b*. In addition, the cap cover **60** may be independently disposed on remaining electrode terminals **1352***a* and **1352***b* to cover the remaining electrode terminals **1352***a* and **1352***b*.

[0084] That is, one of the first terminal portion **135***a* and the second terminal portion **135***b* may be covered by the terminal cap assembly **50**, and the first terminal portion **135***a* and the second terminal portion **135***b* may be electrically connected to each other, and a remaining terminal portion may be independently covered by the cap cover **60**.

[0085] The cap cover **60** may be inserted into the seating space C of the cover support portion **136** to cover the terminal portion **135**.

[0086] Specifically, in at least one of embodiments of the present disclosure, the battery module of the present disclosure may include a plurality of sub-modules **1000** (a first sub-module **100***a* and a second sub-module **100***b*), a first terminal portion **135***a* and a second terminal portion **135***b* provided in the first sub-module **100***a* and the second sub-module **100***b*, and a terminal cap assembly **50** electrically connecting the first terminal portion **135***a* and the second terminal portion **135***b* to each other, the terminal cap assembly **50** covering at least a portion of the first terminal portion **135***a* and the second terminal portion **135***b* such that the at least a portion is not externally exposed. Here, the terminal cap assembly **50** may be detachably coupled to the first terminal portion **135***a* and the second terminal portion **135***b* while covering the fuse wire **51** electrically connecting the first terminal portion **135***a* and the second terminal portion **135***b* to each other, and at least one of the first terminal portion **135***a* or the second terminal portion **135***b* such that the fuse wire **51** and the at least one of the first terminal portion **135***a* or the second terminal portion **135***b* are not externally exposed.

[0087] Hereinafter, the terminal cap assembly **50** of the present disclosure will be described in detail with reference to the drawings.

[0088] FIG. **6** is a perspective view of a terminal cap assembly according to an embodiment of the present disclosure. FIG. **7** is a view of FIG. **6** in a different direction. FIG. **8** is a bottom view of a terminal cap assembly according to an embodiment. FIG. **9** is a side view of a terminal cap assembly according to an embodiment.

[0089] Referring to FIGS. **6** to **9** together, the terminal cap assembly **50** of the present disclosure may include a fuse wire **51** electrically connecting terminal portions **135** of a plurality of adjacent sub-modules **100** to each other, and a protector **53** covering the fuse wire **51** and the terminal portion **135**.

[0090] The fuse wire **51** may electrically connect the terminal portions **135** of the adjacent sub-modules **100** to each other, and may fracture itself when a predetermined current flows. For example, the fuse wire **51** may fracture when a predetermined current flows due to a small thickness thereof.

[0091] Specifically, the fuse wire **51** may have a predetermined diameter, enabling the fuse wire **51** to be melted by heat caused by an overcurrent and to rapidly fracture within one second. For example, the fuse wire **51** may have a diameter of 25 mm.sup.2 to 1.00 mm.sup.2, such that the fuse wire **51** may fracture within one second due to a predetermined overcurrent.

[0092] However, the above-described diameter of the fuse wire **51** is only an example, and it may be understood that any diameter of the fuse wire **51**, enabling the fuse wire **51** to fracture itself when an overcurrent flows and to cut off an electrical connection between the sub-modules **100**, is within the scope of the present disclosure. For example, the fuse wire **51** may be provided as a plurality of wires having different diameters, and the plurality of wires may be provided to fracture according to a degree of overcurrent.

[0093] That is, at least a portion the fuse wire **51** of the present disclosure may be surrounded by the protector **53**, and thus may electrically connect the first terminal portion **135***a* and the second terminal portion **135***b* to each other.

[0094] The fuse wire **51** may include a connection portion **511** in contact with and electrically connected to a plurality of terminal portions **135**, and a connector **513** connecting a plurality of connection portions **511** to each other, the connector **513** that may be covered by the protector **53**.

[0095] The connection portion **511** may include a first connection portion **511** in contact with the first terminal portion **135***a*, and a second connection portion **135***b* in contact with the second terminal portion **135***b*, and the first connection portion **511** and the second connection portion **135***b* may have through-holes, such that protruding bodies **5312***a* and **5312***b* of the protector **53** to be described below may be inserted into the through-holes. In addition, the first connection portion **511***a* and the second connection portion **511***b* may be inserted into fixing bodies **5314***a* and **5314***b*, respectively, and fixed to the protector **53**.

[0096] As described above, the connector **513** may have a diameter, enabling the connector **513** to fracture when an overcurrent flows. However, the present disclosure is not limited thereto, and the connection portion **511** may also fracture.

[0097] The protector **53** may include a plurality of terminal covers **531** covering a plurality of adjacent terminal portions **135**, and a connection cover **535** connecting the plurality of terminal covers **531**. The protector **53** may cover the terminal portion **135** as well as the fuse wire **51** such that the terminal portion **135** as well as the fuse wire **51** are not externally exposed.

[0098] Specifically, the terminal cap assembly **50** may include a terminal cover **531** covering terminal portions **135** of different sub-modules **100**, and a connection cover **535** disposed between the terminal portions **135**, the connection cover **535** surrounding the connector **513** of the fuse wire **51**.

[0099] The terminal cover **531** may include a first terminal cover **531***a* covering one of electrode terminals of the first terminal portion **135***a*, and a second terminal cover **531***b* covering one of electrode terminals of the second terminal portion **135***b*.

[0100] For example, the first terminal cover **531***a* may include a first cover body **5311***a* covering a first electrode terminal **1351***a* of the first terminal portion **135***a*, and a first protruding body **5312***a* protruding from the first cover body **5311***a*, the first protruding body **5312***a* inserted into a terminal hole h of the first terminal portion **135***a*. In addition, the first terminal cover **531***a* may include a first seating body **5313***a* supporting the first connection portion **511***a* of the fuse wire **51** in the first protruding body **5312***a*, a first fixing body **5314***a* fixing the first connection portion **511***a*, and a first support body **5316***a* having a wire hole **5315** through which the fuse wire **51** passes.

[0101] For example, the second terminal cover **531***b* may include a second cover body **5311***b* covering a first electrode terminal **1351***b* of the second terminal portion **135***b*, and a second protruding body **5312***b* protruding from the second cover body **5311***b*, the second protruding body **5312***b* inserted into a terminal hole h of the second terminal portion **135***b*. In addition, the second terminal cover **531***b* may include a second seating body **5313***b* supporting the second connection portion **511***b* of the fuse wire **51** in the second protruding body **5312***b*, a second fixing body **5314***b* fixing the second connection portion **511***b*, and a second support body **5316***b* having a wire hole **5315** through which the fuse wire **51** passes.

[0102] Respective configurations of the first terminal cover **531***a* and the second terminal cover **531***b* described above may be only different from each other in that the first terminal cover **531***a* and the second terminal cover **531***b* are disposed on different terminals **135**, and may be the same configuration. Accordingly, the respective configurations of the first terminal cover **531***a* and the second terminal cover **531***b* will be described together.

[0103] Cover bodies **5311***a* and **5311***b* and support bodies **5316***a* and **5316***b* may cover a seating space C such that the terminal portions **135***a* and **135***b* are not externally exposed (see FIGS. **4** and **5**). That is, the terminal portions **135***a* and **135***b* may be covered by cover support portions **136***a* and **136***b*, the cover bodies **5311***a* and **5311***b*, and the support bodies **5316***a* and **5316***b* not to be externally exposed.

[0104] The protruding bodies **5312***a* and **5312***b* may protrude from the cover bodies **5311***a* and **5311***b* in a predetermined direction (for example, a direction toward a step portion **135**). At least portions of the protruding bodies **5312***a* and **5312***b* may be inserted into and fixed to the through-holes **135***h* of the terminal portions **135***a* and **135***b*, respectively. In addition, as described above, the protruding bodies **5312***a* and **5312***b* may pass through the through-holes of the connection portions **511***a* and **511***b* together.

[0105] That is, the protruding bodies **5312***a* and **5312***b* may sequentially pass through the through-holes of the connection portions **511***a* and **511***b* and the through-holes **135***b* of the terminal portions **135***a* and **135***b*, and may be fixed to the adjacent sub-modules **100***a* and **100***b*.

[0106] Referring to FIG. **7**, seating bodies **5313***a* and **5313***b* may be formed at one sides of the protruding bodies **5312***a* and **5312***b*, such that the connection portions **511***a* and **511***b* may be seated on the terminal portions **135***a* and **135***b*. Accordingly, the protruding bodies **5312***a* and **5312***b* may pass through the connection portions **511***a* and **511***b*, such that the connection portions **511***a* and **511***b* may be stably disposed between the seating bodies **5313***a* and **5313***b* and the terminal portions **135***a* and **135***b*.

[0107] In addition, the cover bodies **5311***a* and **5311***b* may oppose the terminal portions **135***a* and **135***b* in a first direction (a Z-axis direction in the drawings), and the support bodies **5316***a* and **5316***b* may extend in the first direction from one sides of the cover bodies **5311***a* and **5311***b*, and may oppose the terminal portions **135***a* and **135***b* in a direction (a Y-axis direction in the drawings), perpendicular to the first direction.

[0108] Such a structure may allow the terminal portions **135***a* and **135***b* not to be externally exposed by the cover portions **136***a* and **136***b*, the cover bodies **5311***a* and **5311***b*, and the support bodies **5316***a* and **5316***b*.

[0109] The seating bodies **5313***a* and **5313***b* may support the connection portions **511***a* and **511***b* to be in close contact with the terminal portions **135***a* and **135***b* in a state in which the terminal cap assembly **50** is mounted on the terminal portions **135***a* and **135***b*. Accordingly, even when vibrations or impacts that may occur during transportation of the battery module **10** are applied, a state in which the connection portions **511***a* and **511***b* are in contact with the terminal portions **135***a* and **135***b* may be maintained.

[0110] The fixing bodies **5314***a* and **5314***b* may fix the fuse wire **51** to predetermined positions in the cover bodies **5311***a* and **5311***b*.

[0111] The connection cover **535** may accommodate the fuse wire **51** such that the fuse wire **51** is not externally exposed, and may connect the first terminal cover **531***a* and the second terminal cover **531***b* to each other. The connection cover **535** may surround at least a portion of the connector **513** to prevent the connector **513** from being in contact with an adjacent member, such as a module case **11**.

[0112] As described, referring to FIGS. **4** and **5** together, the terminal portions **135***a* and **135***b* may be electrically connected to each other by the fuse wire **51**, and may not be externally exposed by the protector **53**.

[0113] Accordingly, the terminal cap assembly **50** of the present disclosure may cut off an electrical connection between the plurality of sub-modules **100** when an overcurrent of the plurality of sub-modules **100** occurs, and may cover the terminal portion **135** such that the terminal portion **135** is not externally exposed. In addition, the terminal cap assembly **50** of the present disclosure may minimize degradation in durability of the fuse wire **51** due to vibrations or impacts occurring during transportation through structures of the connection cover **535** and the terminal covers **531***a* and **531***b*.

[0114] According to another embodiment of the present disclosure, the terminal cap assembly of the present disclosure may be provided as a plurality of terminal cap assemblies **50** in the battery module **10**. For example, in FIG. **2**, it is illustrated that one terminal cap assembly **50** is provided and disposed only at one side of the battery module **10**, but two terminal cap assemblies **50** may be provided and disposed on both sides of the battery module **10** depending on an electrical connection relationship between the sub-modules **100**.

[0115] That is, when the above-described terminal cap assembly **50** is provided as at least one terminal cap assembly **50**, it may be understood that both one terminal cap assembly **50** and two terminal cap assemblies **50** are within the scope of the present disclosure.

[0116] In addition, the terminal cap assembly **50** may be provided to be detachable from the terminal portion **135**.

[0117] After stable transportation through the terminal cap assembly **50** is completed, an operator may remove the terminal cap assembly **50** and connect various electric devices such as electric vehicles and the like to the terminal portion **135**.

[0118] Only specific examples of implementations of certain embodiments are described. Variations, improvements and enhancements of the disclosed embodiments and other embodiments may be made based on the disclosure of this patent document.

[0119] The features described above are merely examples of the application of the principles of the present disclosure, and other components may be included without departing from the scope of the present disclosure.

Claims

1. A battery module comprising: a plurality of sub-modules including a first sub-module and a second sub-module including a plurality of battery cells, respectively; a first terminal portion electrically connected to the plurality of battery cells of the first sub-module, the first terminal portion including one or more electrode terminals including a first electrode terminal; a second terminal portion electrically connected to the plurality of battery cells of the second module, the second terminal portion including one or more electrode terminals including a first electrode terminal; and a terminal cap assembly covering the first terminal portion and the second terminal portion, the terminal cap assembly electrically connecting the first terminal portion and the second terminal portion to each other, wherein the terminal cap assembly includes: a fuse wire electrically connecting the first electrode terminal of the first terminal and the first electrode terminal of the second terminal to each other; and a protector covering the fuse wire, the first electrode terminal of

the first terminal, and the first electrode terminal of the second terminal.
2. The battery module of claim 1, further includes a cap cover covering the electrode terminal that is not covered by the terminal cap assembly, wherein the plurality of battery cells are stacked in a first direction.
3. The battery module of claim 2, wherein the first terminal portion further includes a second electrode terminal spaced apart from the first electrode terminal of the first terminal portion in the first direction, and the second terminal portion further includes a second electrode terminal spaced apart from the first electrode terminal of the second terminal portion in the first direction, and the cap cover independently covers the second electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion.
4. The battery module of claim 3, wherein the first electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion have different electrical polarities, and a terminal cap assembly is disposed at one sides of the plurality of sub-modules in the first direction, and a cap cover is disposed at the other sides of the plurality of sub-modules in the first direction.
5. The battery module of claim 3, wherein the terminal cap assembly is provided as one terminal cap assembly and connected to the first electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion, and the terminal cap is provided as two terminal caps, and the two terminal caps are disposed on the second electrode terminal of the first terminal portion and the second electrode terminal of the second terminal portion, respectively.
6. The battery module of claim 1, wherein the fuse wire is provided to fracture due to a current having a predetermined value or more, and to cut off an electrical connection between the first terminal portion and the second terminal portion.
7. The battery module of claim 6, wherein the fuse wire has a diameter of 0.25 mm.sup.2 to 1 mm.sup.2.
8. The battery module of claim 1, wherein the protector includes: a first terminal cover covering the first electrode terminal of the first terminal portion; a second terminal cover covering the first electrode terminal of the second terminal portion; and a connection cover connecting the first terminal cover and the second terminal cover to each other.
9. The battery module of claim 8, wherein the electrode terminal of the first terminal portion and the electrode terminal of the second terminal portion has terminal holes, respectively, and the first terminal cover and the second terminal cover include a first protruding body and a second protruding body protruding to be inserted into the terminal holes, respectively.
10. The battery module of claim 9, wherein the fuse wire includes a first connection portion and a second connection portion in contact with and electrically connected to the first electrode terminal of the first terminal portion and the first electrode terminal of the second terminal portion, respectively, and the first protruding body and the second protruding body pass through the first connection portion and the second connection portion, and are inserted into the first electrode terminal of the first terminal portion and the first electrode terminal of the second terminal portion, respectively.
11. The battery module of claim 10, wherein the protector includes a first seating body and a second seating body respectively provided at one sides of the first protruding body and the second protruding body, and in a state in which the terminal cap assembly is coupled to the first terminal portion and the second terminal portion, the first connection portion is disposed between the first seating body and the first terminal portion, and the second connection portion is disposed between the second seating body and the second terminal portion.
12. The battery module of claim 11, wherein the fuse wire further includes a connector connecting the first connection portion and the second connection portion to each other, and at least a portion of the connector is inserted into the connection cover and is not externally exposed.
13. The battery module of claim 9, further comprising: a first cover support portion and a second

cover support portion protruding from edges of the first and second terminal portions, the first cover support portion and the second cover support portion forming a seating space in which the terminal cap assembly is to be mounted, wherein the terminal cap assembly is mounted in the seating space to cover the first electrode terminal of the first terminal portion and the first electrode terminal of the second terminal portion.

14. The battery module of claim 13, wherein the first terminal cover and the second terminal cover respectively include: a first cover body and a second cover body disposed to respectively oppose the first terminal portion and the second terminal portion in the first direction; and a first support body and a second support body respectively extending from one sides of the first cover body and the second cover body in the first direction, the first support body and the second support body respectively opposing the first terminal portion and the second terminal portion in a second direction, perpendicular to the first direction.

15. A battery module comprising: a plurality of sub-modules including a first sub-module and a second sub-module respectively including a plurality of battery cells stacked in a first direction; a first terminal portion electrically connected to the plurality of battery cells of the first sub-module, the first terminal portion including a plurality of electrode terminals; a second terminal portion electrically connected to the plurality of battery cells of the second sub-module, the second terminal portion including a plurality of electrode terminals; a terminal cap assembly electrically connecting one electrode terminal, among the plurality of electrode terminals of the first terminal portion and one electrode terminal, among the plurality of electrode terminals of the second terminal portion, to each other, the terminal cap assembly covering the one electrode terminal of the first terminal portion and the one terminal of the second terminal portion together; and a plurality of cap covers independently covering remaining electrode terminals, among the plurality of electrode terminals of the first terminal portion, and remaining electrode terminals, among the plurality of electrode terminals of the second terminal portion, respectively, wherein the terminal cap assembly is disposed at one side of the first and the second sub-modules in the first direction, and the plurality of cap covers are disposed at the other side of the first and the second sub-modules in the first direction.

16. A terminal cap assembly comprising: a fuse wire electrically connecting a first terminal portion and a second terminal portion, respectively,; and a protector covering at least a portion of the fuse wire, the first terminal portion, and the second terminal portion, the protector coupled to the first terminal portion and the second terminal portion.

17. The terminal cap assembly of claim 16, wherein the fuse wire is provided between the first terminal portion and the second terminal portion to fracture due to a current having a predetermined value or more, and to cut off an electrical connection between the first terminal portion and the second terminal portion.

18. The terminal cap assembly of claim 17, wherein the fuse wire includes: a first connection portion and a second connection portion respectively in contact with the first terminal portion and the second terminal portion; and a connector electrically connecting the first connection portion and the second connection portion to each other.

19. The terminal cap assembly of claim 18, wherein the protector includes: a first terminal cover and a second terminal cover respectively covering the first terminal portion and the second terminal portion; and a connection cover disposed between the first terminal cover and the second terminal cover, the connection cover surrounding the connector.

20. The terminal cap assembly of claim 19, wherein the first terminal cover and the second terminal cover respectively include a first protruding body and a second protruding body respectively passing through the first connection portion and second connection portion, the first protruding body and the second protruding body respectively inserted into through-holes of the first terminal portion and the second terminal portion.